

Use Case “Update Product”

1. Use case code

UC234.2

2. Brief Description

This use case allows the Product Manager to update information for a single media product in the AIMS system at a time. It ensures the current price remains between 30% and 150% of the original value, with both figures excluding 10% VAT. Depending on the category, the system validates specific mandatory fields such as authors, publishers, or directors. The software records all changes in the history log and notifies the user of any invalid data or rule violations.

3. Actors

3.1 Product Manager

4. Preconditions

- Product Manager successfully logged into the AIMS System with appropriate account provided.
- The AIMS System is currently operating normally.

5. Basic Flow of Events

1. Product Manager request access to the product management interface.
2. Software displays list of products and options for adding, editing, deleting and adjusting quantities.
3. Product Manager select one product
4. Product Manager select Edit Function
5. Software displays product’s current information
6. Product Manager update the necessary information
7. Product Manager submit the form
8. Software check constraint of data and price
9. Software update product information

10. Software display notifications and save record into history.

6. Alternative flows

Table 1 - Alternative flows of events for UC “Update Product”

No	Location	Condition	Action	Resume location
1.	At step 8	If the input data is not in the correct format	▪ Software display detailed error for each data field	At step 6
2.	Any step	If Product Manager cancel process	▪ Software do not save any changes and return product list interface	At step 3

7. Input data

Table 2 - Input data of UC “Update Product”

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Title	The name of the media product	Yes	Cannot be blank	DVD Hello World
2.	Category	Product classification	Yes	One of: Books, Newspapers, CDs, DVDs	Books
3.	General Description	Condition, primary color, or return condition	Yes	Text description	New condition, blue cover
4.	Original Value	Base price excluding 10% VAT	Yes	Positive integer	100
5.	Current Price	Selling price excluding 10% VAT	Yes	Between 30% and 150% of Original Value	120
6.	Dimensions	Physical size (Height x	Yes	Positive values only	20 x 15 x 3 (cm)

		Width x Length)			
7.	Weight	Physical weight of the product	Yes	Positive value (kg)	0.5
8.	Author	Creator(s) of the book	Yes (Book)	Required if Category is "Books"	Ngo Tat To
9.	Cover Type	Format of the book cover	Yes (Book)	"Paperback" or "Hardcover"	Paperback
10.	Editor-in-chief	Head of the newspaper	Yes (News)	Required if Category is "Newspapers"	Nguyen Van A
11.	Artists	Performers or singers	Yes (CD)	Required if Category is "CDs"	Son Tung M-TP
12.	Track List	List of songs with titles and lengths	Yes (CD)	Must include title and duration per track	Track 1 - 10:30
13.	Disc Type	Physical format of the disc	Yes (DVD)	"Blu-ray" or "HD-DVD"	Blu-ray
14.	Director	Director of the media	Yes (DVD)	Required if Category is "DVDs"	Christopher Nolan

8. Output data

Table 3 - Output data of UC “Update Product”

No	Data fields	Description	Display format	Example
1.	Operation Status	Notification of the result for the performed action (add, edit, delete, or adjust).	Text notification	Product updated successfully
2.	Error Message	Detailed notification when input data is invalid or violates system rules.	Text (Red)	Price must be between 30% and 150% of Original Value
3.	History Log	A record list of all product additions, edits, and deletions.	Timeline/Log entry	16/03/2026: Manager A updated information of Barcode X

9. Postconditions

- Data Persistence: Product information is accurately updated to reflect the latest version in the database.
- Error message: Product Manager receive error text message at invalid field.
- Interface Refresh: The product management interface is automatically refreshed to display the most recent data after the operation.
- Audit trail: A new record created in history, log the detail information of the operation
- User notification: Product Manager receive notification from software, product list interface must refreshed to display updated list